

Test and Reliability

Exercise Sheet 5

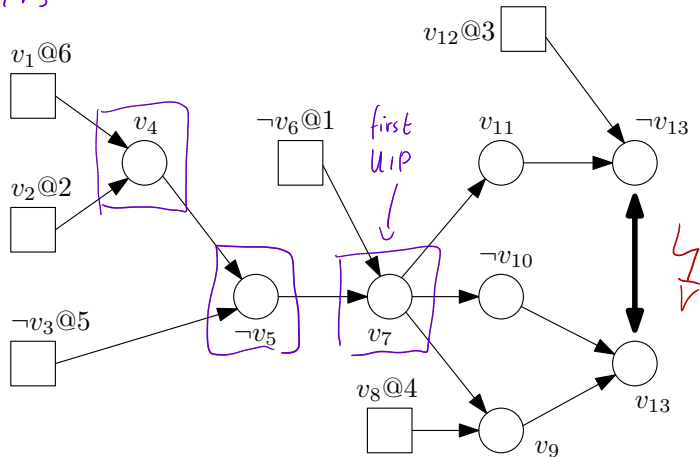
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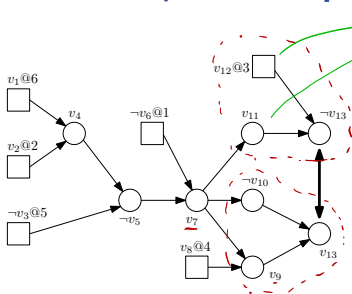


Exercise 1) First Unique Implication Point (1UIP)

UIPs:



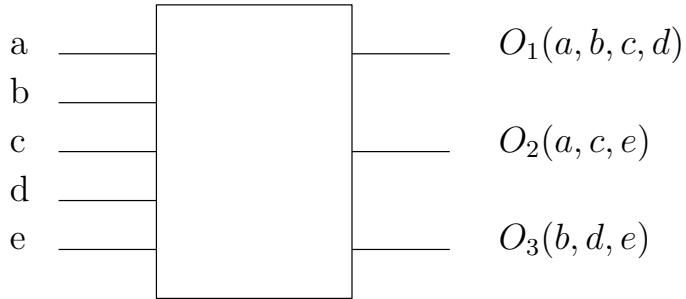
Exercise 1) First Unique Implication Point (1UIP)



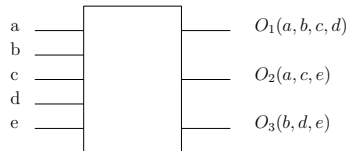
$$\begin{aligned}
 & (\neg v_{12} \vee \neg v_{11} \vee \neg v_{13}) \wedge (\underbrace{v_{10} \vee \neg v_9 \vee \neg v_{13}}_{c_2}) \wedge \\
 & (\neg v_7 \vee \underbrace{v_{11}}_{c_3}) \wedge (\neg v_7 \vee \neg v_{10} \vee \neg v_9) \wedge (\neg v_7 \vee \neg v_8 \vee \neg v_9) \wedge \\
 & (v_5 \vee v_6 \vee v_7) \wedge (\neg v_4 \vee v_3 \vee \neg v_5) \wedge (\neg v_1 \vee \neg v_2 \vee v_4)
 \end{aligned}$$

$$\begin{aligned}
 \text{Resolve } c_1 \otimes_{v_{13}} c_2 &= (\neg v_{12} \vee \neg v_{11} \vee v_{10} \vee \neg v_9) = c_{1,2} \\
 c_{1,2} \otimes_{v_{11}} c_3 &= (\neg v_{12} \vee v_{10} \vee \neg v_9 \vee \neg v_7) = c_{1,2,3} \\
 c_{1,2,3} \otimes_{v_{10}} c_4 &= (\neg v_{12} \vee \neg v_9 \vee \neg v_7) = c_{1,2,3,4} \\
 c_{1,2,3,4} \otimes_{v_9} c_5 &= (\neg v_{12} \vee \neg v_7 \vee \neg v_8)
 \end{aligned}$$

Exercise 2) Pseudo-exhaustive Testing



Exercise 2a) Pseudo-exhaustive Testing



$$\underline{I} = 5 \quad p = 5 \quad w = 4$$

$$\underline{w \leq p \leq I}$$

	a	b	c	d	e
O_1	1	1	1	1	0
O_2	1	0	1	0	1
O_3	0	1	0	1	1

→ nothing

pairwise
compatible

→
we need as
many test signal
inputs as there
are circuit PIs

Exercise 2b) Pseudo-exhaustive Testing

Minimal example with $I = O = 3$, $w = 2$, $p = 3$. Obviously, $w < p$.

	a	b	c
O_1	1	1	0
O_2	1	0	1
O_3	0	1	1

3.

	a	b	c
O_1	1	0	0
O_2	x_1	x_2	x_3

 $\Rightarrow x_2 = x_3 = 1$
(because otherwise b,c disconnected)
 $\Rightarrow w \geq 2$
If $x_1 = 1$, then Case 1. again.
If $x_1 = 0$, then a,b and a,c are compatible $\rightarrow p = w = 2$

To do: Is it really minimal? Proof:

First try: reduce I . If $I < 3$, then $p < 3$, then $w < p < 3$, thus $w = 1$. Which, in turn, would mean that also $p = 1$.

Second try: reduce O . (leave $I = 3$). Try $O = 1$, then it depends on all inputs (otherwise at least one input would be disconnected) $\rightarrow w = p = I = 3$

Try $O = 2$: case distinction (note: $x_{1,2,3}$ are either 0 or 1)

1.

	a	b	c
O_1	1	1	1
O_2	x_1	x_2	x_3

 $\Rightarrow w = 3$ and $I = 3$ $\Rightarrow w = p = 3$

2.

	a	b	c
O_1	1	1	0
O_2	x_1	x_2	x_3

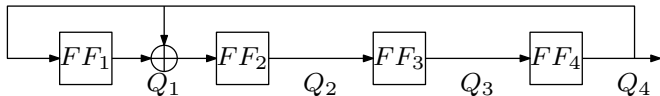
 $\Rightarrow x_3 = 1$ and either $x_1 = 0$ or $x_2 = 0$ because otherwise: case 1. $\rightarrow p = w = 2$

Exercise 3) Linear Feedback Shift Registers

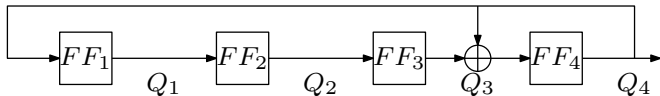
$$x^4 + x + 1$$



Option 1:



Option 2:



$$x^4 + x^3 + 1$$


Exercise 3) Linear Feedback Shift Registers

max. cycle length

$$2^n - 1$$

$$\rightarrow 2^4 - 1 = 15$$

LFSR 1					LFSR 2			
Q ₁	Q ₂	Q ₃	Q ₄		Q ₁	Q ₂	Q ₃	Q ₄
0	0	0	1		0	0	0	1
1	1	0	0		1	0	0	1
0	1	1	0		1	1	0	1
0	0	1	1		1	1	1	1
1	1	0	1		1	1	1	0
1	0	1	0		0	1	1	1
0	1	0	1		1	0	1	0
1	1	1	0		0	1	0	1
0	1	1	1		1	0	1	1
1	1	1	1		1	1	0	0
1	0	1	1		0	1	1	0
1	0	0	1		0	0	1	1
1	0	0	0		1	0	0	0
0	1	0	0		0	1	0	0
0	0	1	0		0	0	1	0
0	0	0	1		0	0	0	1

assumption: seeded
into (0, 0, 0, 1)

Exercise 4) Polynomials

a) $x^2 + 1 = x^2 - 1$
 $\Downarrow$
 not reducible in general, but in $GF(2)$ in $GF(2)$

$$x^2 - 1 = (x + 1) \cdot (x - 1)$$



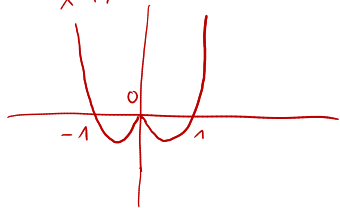
b) degree 3: $\cancel{x^3}$
 $\cancel{x^3 + x^2}$ } root $x = 0$
 $\cancel{x^3 + x^2 + x}$

irreducible because we can not split off a linear factor
 $\left\{ \begin{array}{l} \cancel{x^3 + x^2 + x + 1} \\ x^3 + x^2 + x + 1 \\ x^3 + x + 1 \end{array} \right.$
 $\rightarrow$ root $x = 1$
 (no roots), they would have to be factored by irreducible quadratic polynomials $\rightarrow$ does not work with degree of 3

not relevant for the exam!

Exercise 4) Polynomials

c) GCD (greatest common divisor) of $x^4 + x^2$ and $x^4 + 1$



$$x^4 + x^2 = x(x-0)(x+1)(x-1)$$



$$x^4 + 1 = \overbrace{(x^2 - 1)}^{x^2 - 1} (x+1)(x-1)$$

$\Rightarrow x^2 - 1$ is the GCD

not relevant
for the final
exam